

Arquivo: run_cluster.py

```
from __future__ import annotations

import os
import subprocess
import sys
import time
from pathlib import Path

ROOT_DIR = Path(__file__).resolve().parent
DFS_DIR = ROOT_DIR / "DFS_M2"
SRC_DIR = DFS_DIR / "src"

def build_env() -> dict[str, str]:
    """
    Monta o ambiente para que os subprocessos encontrem o pacote dfs.
    """
    env = os.environ.copy()
    current_pythonpath = env.get("PYTHONPATH", "")

    paths = [str(SRC_DIR)]
    if current_pythonpath:
        paths.append(current_pythonpath)

    env["PYTHONPATH"] = os.pathsep.join(paths)
    return env

def start_process(label: str, args: list[str], cwd: Path, env: dict[str, str]) -> subprocess.Popen:
    """
    Inicia um processo filho e devolve o handle dele.
    """
    print(f"[INICIANDO] {label}: {' '.join(args)}")
    return subprocess.Popen(
        args,
        cwd=str(cwd),
        env=env,
    )

def main() -> None:
    """
    Sobe os três nós e o coordenador do DFS.
    """
    if not DFS_DIR.exists():
        print(f"Erro: pasta não encontrada: {DFS_DIR}")
        sys.exit(1)

    env = build_env()
    processes: list[subprocess.Popen] = []
```

```

try:
    processes.append(
        start_process(
            "node1",
            [sys.executable, "-m", "dfs.interface.storage_node", "--node-id", "node1"],
            cwd=DFS_DIR,
            env=env,
        )
    )
    time.sleep(0.5)

    processes.append(
        start_process(
            "node2",
            [sys.executable, "-m", "dfs.interface.storage_node", "--node-id", "node2"],
            cwd=DFS_DIR,
            env=env,
        )
    )
    time.sleep(0.5)

    processes.append(
        start_process(
            "node3",
            [sys.executable, "-m", "dfs.interface.storage_node", "--node-id", "node3"],
            cwd=DFS_DIR,
            env=env,
        )
    )
    time.sleep(0.5)

    processes.append(
        start_process(
            "coordinator",
            [sys.executable, "-m", "dfs.interface.server"],
            cwd=DFS_DIR,
            env=env,
        )
    )

    print("\nCluster DFS iniciado.")
    print("Deixe este terminal aberto enquanto usa a CLI em outro terminal.")
    print("Para encerrar, pressione Ctrl+C.\n")

    while True:
        time.sleep(1)

except KeyboardInterrupt:
    print("\nEncerrando cluster...")

finally:

```

```
for proc in processes:
    if proc.poll() is None:
        proc.terminate()

for proc in processes:
    try:
        proc.wait(timeout=3)
    except subprocess.TimeoutExpired:
        proc.kill()

print("Cluster encerrado.")

if __name__ == "__main__":
    main()
```